STUDENT DATA ANALYSIS REPORT

1.Data Understanding

- Total Records: 561
- Total Columns: 12
- Column Names: Student ID, Student Names, Course, Semesters 1-6, Average Marks, Grade, Status, Attendance.

  - Who are the students who have scored above 90% of the average marks?
  - What is the average marks of each course?
  - How many students scored above 80% ?
  - Which course has the lowest academic performance?
  - How many students have attendance below 75%?
  - Does low attendance affect pass/fail status?
  - How does attendance influence overall grade distribution?
  - Is there a trend of improvement across semesters?

2.Data Cleaning

- Checked for missing values and handled them appropriately.
- Removed duplicate records if any.
- Standardized column names and data formats.
- Ensured numerical columns are in correct datatype.

3.Business KPI

Key Performance Indicators (KPIs) are measurable values used to evaluate student academic performance and overall progress, Like the effectiveness of teaching , subject wise curriculum difficulty analysis, academic success rate, learning progress of student, performance analysis and what should be improved.

4.KPI List

- Average Marks per Semester
- Overall Percentage
- Pass/Fail Rate
- Highest Scoring Student

- Lowest Scoring Student
- Highest Attendance%
- Lowest Attendance%
- High Distinction

5.Business Insights

• Identify top-performing students for recognition.

• Detect students at academic risk based on low averages.

• Analyze semester-wise trends to improve curriculum.

• Support data-driven decision making for academic planning.

- Corerlation attendance with marks enables early intervention for struggling students.